A Hierarchical Geometry-guided Transformer for Histological Subtyping of Primary Liver Cancer

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***Abstract*—Primary liver malignancies are widely recognized as the most heterogeneous and prognostically diverse cancers of the digestive system. Among these, hepatocellular carcinoma (HCC) and intrahepatic cholangiocarcinoma (ICC) emerge as the two principal histological subtypes, demonstrating significantly greater complexity in tissue morphology and cellular architecture than other common tumors. The intricate representation of features in Whole Slide Images (WSIs) encompasses abundant crucial information for liver cancer histological subtyping, regarding hierarchical pyramid structure, tumor microenvironment (TME), and geometric representation. However, recent approaches have not adequately exploited these indispensable effective descriptors, resulting in a limited understanding of histological representation and suboptimal subtyping performance. To mitigate these limitations, A hieRarchical Geometry-gUided tranSformer (ARGUS) is proposed to advance histological subtyping in liver cancer by capturing the macro-meso-micro hierarchical information within the TME. Extensive experiments on public and private cohorts demonstrate that our ARGUS achieves state-of-the-art (SOTA) performance in histological subtyping of liver cancer, which provide an effective diagnostic tool for primary liver malignancies in clinical practice. Related code will be available to public.**

***Index Terms*—Computational Pathology, Histological Subtyping, Weakly-Supervised Learning, Geometric Representation.**

I. Introduction

Primary liver cancer is the fourth leading cause of cancer-related mortality worldwide. Its two main subtypes, hepatocellular carcinoma (HCC) and intrahepatic cholangiocarcinoma (ICC), arise from hepatocytes and biliary epithelial cells, respectively, and exhibit distinct histopathology and clinical behavior. ICC is highly aggressive and heterogeneous, with poorer prognosis and greater histological complexity than HCC, making accurate subtyping clinically important for personalized treatment.

Clinically, ICC is classified into large, small, and fine duct types based on biliary origin and histopathology [7]. Large duct ICC is aggressive with poor outcomes, while fine duct ICC shows lower invasiveness and better prognosis. Recently, some studies provided primary liver cancer diagnostic solutions using radiology or histology images for HCC vs. ICC or HCC fine-grained subtyping. However, few studies have investigated the subtyping of ICC using WSIs, which exhibit high inter-subtype similarity and thus render this a challenging fine-grained classification task.

Histological subtyping of primary liver cancer is a challenging task, requiring both coarse-grained features, such as tumor size/invasion, lymphocytic infiltrates, and the broad organization of these phenotypes in the TME, also fine-grained features such as nuclear atypia or tumor presence, for assessing precise subtyping of malignancy [3]. Recent approaches in similar tasks almost adapt the multiple instance learning (MIL) framework [1], [10], [12], [13], [16], which are unable to capture important contextual and hierarchical information in the cancer diagnosis [5]. To this end, multi-scale/FoV and graph-based models have been proposed to address these limitations. For instance, DSMIL [10] leverages tissue features ranging from millimeter-scale to cellular scale, and HIPT [3] learns hierarchical WSI structures via self-supervised learning. However, these methods are not fully context-aware and are unable to model critical morphological feature interactions between cell/nuclei identities and tissue types. Therefore, many graph-based models were presented to capture fine-grained cell-to-cell interactions [14], [20], but typically rely on a coarse patch-based graph convolutional network (GCN), neglecting the rich information from shape, and other useful features of nuclei/cell identities.

In this paper, we propose a graph-based, weakly-supervised framework, dubbed **A** hie**R**archical **G**eometry-g**U**ided tran**S**former (ARGUS), as shown in Fig. 1, tailored for liver cancer histological subtyping by modeling hierarchical interactions across macro-meso-micro resolutions of pathological WSI. The **main contributions** of this paper are as follows:

1) We represent the deepest FoV of WSIs via a micro geometric structure across nuclei identities using hand-crafted features, thereby providing a more fine-grained perspective for pathological image interpretation.
2) We designed a Geometry Prior Guided Fusion strategy (GPGF) to integrate FoVs morphological and geometric features to provide comprehensive learning of WSIs.
3) We performed extensive experiments on two datasets from The Cancer Genome Atlas (TCGA) and in-house cohort, the results demonstrate that our method consistently outperforms current state-of-the-art methods.

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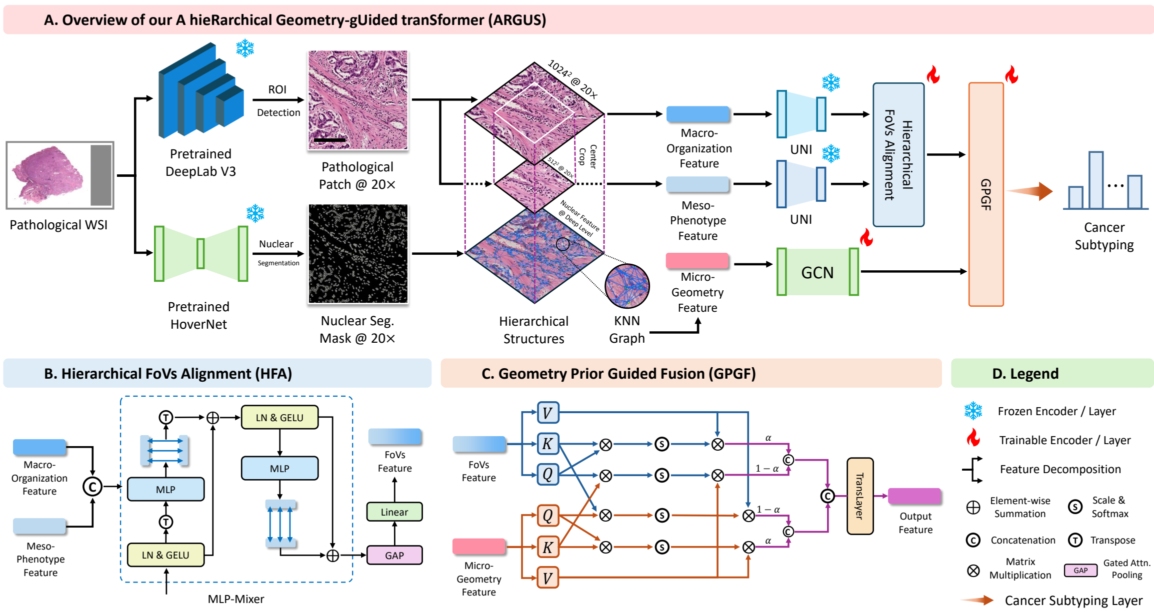

Fig. 1. Overview of the proposed ARGUS. (a) Overall workflow of our framework for histological subtyping, (b) Hierarchical FoVs Alignment (HFA) module, (c) Geometry Prior Guided Fusion (GPGF) module, (d) Legend for the symbols used.

II. Methodology

A. Data Preprocessing and Feature Extraction

1) Data Preprocessing: We pretrained a DeepLabv3 tumor segmentation model [2] to exclude benign and non-tissue regions. WSIs were processed with a sliding window of 562×562 µm and downsampled to 256×256 pixels for pixel-wise tumor classification. Patches with $>$ 50% tumor area were retained at 1024×1024 pixels (0.549 µm/pixel).

2) Histological Feature Extraction: To alleviate the input resolution constraints of pretrained pathological feature extractors, we follow [6] to introduce the hierarchical UNI (hi-UNI) for FoVs histological feature extraction. The original 1024×1024 tumor patch was downsampled to 224×224 (2.509 µm/pixel) to represent macro-level morphological pattern; a 512×512 region which center cropped from original patch was further downsampled to 224×224 to capture meso tissue-level representation. Both FoVs patches are independently processed by UNI [4] to capture global macro-organization (f_{macro}) and local meso-phenotype (f_{meso}) features.

3) Micro-Level Geometric Feature Extraction: To capture micro cellular-level geometric structure within the TME, we leveraged Hover-Net [8] to segment the nuclei and classify them into five categories: tumor, inflammatory, stroma, necrosis (dead) and epithelial (normal). In this work, we focus on tumor, inflammatory, stromal and epithelial nuclei, which represent the major and functionally relevant cellular components for liver cancer. To this end, we extracted three types of handcrafted histological features $\mathbb{X}_i$ for each i-th nucleus: morphological features indicating cell's shape and contour, texture features reflecting local pixel patterns via gray-level co-occurrence matrices (GLCM), and topological features characterizing intercellular relationships. Consequently, we design the micro-geometry feature to represent the micro cellular-level interactions via a geometric structure which can be regarded as the deepest FoV in the pyramid gigapixel WSI. We construct this geometric framework by employing a k-nearest neighbors (k-NN, $k = 8$) algorithm to define edge connectivity, connecting each nucleus to its eight nearest neighbors within a 100-pixel distance threshold. Thus we can build a graph structure via the nucleus and the connectivity across these nucleus, then we follow [20] to implemented a GCN layer to handle the graph structure and further transform it into a micro-geometry feature representation $f_{\mathbf{g}}$.

B. Hierarchical FoVs Alignment Module

As aforementioned, pathological WSIs exhibit a hierarchical pyramid structure across varying resolutions. Therefore, we introduce a Hierarchical FoVs Alignment module (HFA) to capture the hierarchical relationships. Specifically, we leverage a combination of MLP-Mixer and Gated Attention Pooling Network [9], which can enhance hierarchical information interaction as well as produce a contribution-weighted output for multi-FoVs feature aggregation. The detailed operation of our HFA module can be formulated as:

$$\begin{aligned} f_{\text{FoV}} &= \text{HFA}\left(f_{\text{c}}\right), f_{\text{c}} = \text{Concat}\left(f_{\text{macro}}, f_{\text{meso}}\right) \\ \text{HFA}\left(f_{\text{c}}\right) &= \text{Linear}\Big(\text{GAP}(\text{MLP}_{\text{Mixer}}\left(f_{\text{c}}\right))\Big) \end{aligned} \tag{1}$$

Here, $\text{MLP}_{\text{Mixer}}(\cdot)$ and $\text{GAP}(\cdot)$ denote the proposed MLP-Mixer and Gated Attention Pooling. The MLP-Mixer (token- and channel-mixing MLPs) facilitates effective intra- and inter-scale information communication. It can be formulated as:

$$\begin{aligned} \text{Z}_1 &= f_{\text{c}}^{\top} + \text{Linear}(\Theta\left(f_{\text{c}}^{\top}\right) \cdot \text{W}_1) \cdot \text{W}_2 \\ \text{Z} &= \text{Z}_1^{\top} + \text{Linear}(\Theta\left(\text{Z}_1^{\top}\right) \cdot \text{W}_3) \cdot \text{W}_4 \end{aligned} \tag{2}$$

TABLE I
THE PERFORMANCE COMPARED WITH 11 BASELINES ON TWO DATASETS. "M." INDICATES WHETHER TO USE MORPHOLOGICAL FEATURES AND "G." INDICATES WHETHER TO USE GEOMETRY FEATURES. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY.

Methods	Modality		TCGA-Liver				DTH-ICC			
	M.	G.	AUC ↑	ACC ↑	F1 ↑	Pre. ↑	AUC ↑	ACC ↑	F1 ↑	Pre.↑
ABMIL [9]	✓		98.2 ± 1.2	95.9 ± 0.7	87.0 ± 2.0	86.9 ± 4.7	85.2 ± 0.8	69.0 ± 0.3	68.5 ± 1.5	69.0 ± 1.3
DSMIL [10]	✓		97.9 ± 1.9	96.1 ± 1.0	86.8 ± 3.6	88.9 ± 6.6	84.9 ± 0.7	67.7 ± 1.7	67.3 ± 0.7	67.1 ± 1.3
CLAM-SB [15]	✓		98.0 ± 0.6	94.0 ± 1.1	81.7 ± 2.6	77.7 ± 3.2	83.9 ± 1.8	66.5 ± 3.8	67.6 ± 2.4	68.2 ± 2.3
CLAM-MB [15]	✓		98.3 ± 1.3	95.1 ± 1.0	82.0 ± 1.7	88.1 ± 2.5	85.9 ± 0.8	68.4 ± 3.8	68.9 ± 4.4	69.7 ± 3.5
TransMIL [16]	✓		98.1 ± 1.4	95.1 ± 2.1	86.4 ± 3.9	83.2 ± 5.1	83.4 ± 2.5	63.4 ± 4.9	64.1 ± 4.3	67.0 ± 2.3
ACMIL [19]	✓		98.5 ± 1.6	96.3 ± 0.9	87.5 ± 1.9	90.4 ± 1.3	86.2 ± 1.2	69.5 ± 1.2	69.1 ± 1.7	70.0 ± 1.7
IBMIL [11]	✓		98.1 ± 1.5	95.5 ± 1.2	85.1 ± 1.8	88.4 ± 3.3	84.1 ± 1.8	66.5 ± 1.2	67.2 ± 1.9	67.4 ± 1.0
MHIM-MIL [17]	✓		98.7 ± 0.8	96.4 ± 0.5	89.0 ± 2.6	86.8 ± 2.7	86.4 ± 1.5	69.1 ± 2.2	70.0 ± 0.9	69.9 ± 1.1
Patch-GCN [5]		✓	96.2 ± 3.2	94.1 ± 1.9	78.4 ± 3.0	85.2 ± 3.1	74.7 ± 1.5	59.5 ± 2.5	58.2 ± 2.2	60.7 ± 1.2
GTMIL [20]		✓	97.9 ± 1.5	96.4 ± 1.3	85.4 ± 2.0	93.1 ± 2.5	82.6 ± 1.4	61.3 ± 2.1	61.5 ± 1.7	61.4 ± 0.9
NPKC-MIL [18]	✓	✓	99.1 ± 1.1	96.9 ± 0.9	91.8 ± 2.3	91.2 ± 3.7	86.8 ± 2.5	70.2 ± 2.2	70.6 ± 2.3	71.8 ± 1.9
ARGUS (Ours)	✓	✓	99.5 ± 0.3	98.1 ± 0.7	93.5 ± 2.3	93.6 ± 5.3	88.4 ± 1.3	74.0 ± 0.9	73.7 ± 0.6	74.6 ± 0.4

where $\Theta(\cdot)$ denotes the combination of Layer Normalization and GELU, $\mathrm{W}_1, \mathrm{W}_2, \mathrm{W}_3$ and W_4 are trainable weights of the fully connected linear layers. Then, the hidden state Z will be sent into a shared gated attention pooling network $\mathrm{GAP}(\cdot)$, which consists of two linear layers with ReLU and Sigmoid, the formulation of this procedure can be described as:

$$f_{\mathrm{FoV}} = \alpha_1 \cdot \Phi_1(\mathrm{Z}) + \alpha_2 \cdot \Phi_2(\mathrm{Z}), \alpha_i = \mathrm{Sigmoid}(\Phi_i(\mathrm{Z})) \quad (3)$$

where $\Phi(\cdot)$ is a MLP layer with a architecture of Linear-ReLU-Linear-LayerNorm. We here compute the weighted importance scores assigned to each MLP branch via $\mathrm{Sigmoid}(\cdot)$. In this way, we can obtain an importance-weighted dynamic representation which reflecting the actual contributions for the two FoV features unlike normal fixed-scale fusion strategies.

C. Geometry Prior Guided Fusion Module

To learn the fine-grained contextual relationship among all the nucleus within the TME, we propose a novel geometry prior guided attention operation, termed GPGF, aiming to treat the geometric feature as a prior knowledge to guide the FoVs morphological feature integration. We construct a geometry-aware overall FoVs feature by integrating macro-meso FoVs fusion feature f_{FoV} and micro-geometry feature f_{g}, the specific intra- and inter-modality fusion operation as follows:

$$\begin{aligned} f_{\mathrm{FoV}}^{\mathrm{self}} &= \Gamma(f_{\mathrm{FoV}}, f_{\mathrm{FoV}}, f_{\mathrm{FoV}}), f_{\mathrm{FoV}}^{\mathrm{cross}} = \Gamma(f_{g\to\mathrm{FoV}}, f_{\mathrm{FoV}}, f_{\mathrm{FoV}}), \\ f_g^{\mathrm{self}} &= \Gamma(f_{\mathrm{g}}, f_{\mathrm{g}}, f_{\mathrm{g}}), f_{\mathrm{g}}^{\mathrm{cross}} = \Gamma(f_{FoV\to\mathrm{g}}, f_{\mathrm{g}}, f_{\mathrm{g}}), \end{aligned} \quad (4)$$

where $\Gamma(\cdot)$ indicates the Multi-Head Attention (MHA) mechanism, we then perform a gating strategy to compute the balanced representation dynamically as following:

$$\begin{aligned} f'_{\mathrm{FoV}} &= \alpha_{\mathrm{FoV}} \cdot f_{\mathrm{FoV}}^{\mathrm{self}} + (1-\alpha_{\mathrm{FoV}}) \cdot f_{\mathrm{FoV}}^{\mathrm{cross}} \\ f'_{\mathrm{g}} &= \alpha_{\mathrm{g}} \cdot f_{\mathrm{g}}^{\mathrm{self}} + (1-\alpha_{\mathrm{g}}) \cdot f_{\mathrm{g}}^{\mathrm{cross}} \end{aligned} \quad (5)$$

we further combine the enhanced representations f'_{FoV} and f'_{g} then send it into a Transformer Layer for modeling the long-range dependencies within the final representation:

$$f_{\mathrm{out}} = \mathrm{TransLayer}(\mathrm{Concat}(f'_{\mathrm{FoV}}, f'_{\mathrm{g}})) \quad (6)$$

Lastly, a fully-connected layer is employed to produce the final representation f_{out} for histological subtyping of liver cancer.

III. EXPERIMENTS AND RESULTS

A. Datasets

To comprehensively evaluate the efficacy of ARGUS, we used two liver datasets covering public and clinical cohorts. The first, TCGA-Liver, was curated from the TCGA Data Portal[1] and includes 413 WSIs (379 HCC, 34 ICC) for liver cancer subtyping. To further validate ARGUS's generalizability, we incorporated an in-house cohort: DTH-ICC, from Nanjing Drum Tower Hospital, comprising 789 WSIs for fine-grained ICC subtyping (fine-duct: 289 WSIs / 67 patients; small-duct: 241 WSIs / 78 patients; large-duct: 259 WSIs / 115 patients).

B. Comparison with State-of-the-art Methods

We benchmarked ARGUS against 11 representative baselines under identical experimental settings on both public (TCGA-Liver) and in-house (DTH-ICC) cohorts (see Tab. I). ARGUS consistently achieved superior performance across the two cohorts. In particular, it outperforms TransMIL [16], a classic MIL method, indicating that effective histological subtyping is facilitated by modelling hierarchical TME phenotypes rather than relying on single-scale features. ARGUS also surpasses NPKC-MIL [18], a morphological–geometric multimodal approach, underscoring the value of hierarchical pyramid structures and richer geometric representations. Overall, these results demonstrate the advantage of our multi-scale, geometry-aware design for robust histological subtyping.

[1]TCGA: https://portal.gdc.cancer.gov

TABLE II
QUANTITATIVE RESULTS FOR ABLATION STUDY ON IN-HOUSE DTH-ICC COHORT. WE **BOLD** THE HIGHEST PERFORMANCE.

Model	Designs in our model			DTH-ICC		
	HFA	MGF	GPGF	AUC ↑	ACC ↑	F1 ↑
A				85.0 ± 0.9	66.8 ± 3.5	66.9 ± 3.2
B	✓			86.9 ± 1.5	70.9 ± 3.0	70.6 ± 4.1
C		✓		86.6 ± 1.5	69.0 ± 4.3	68.2 ± 4.5
D	✓	✓		87.1 ± 2.4	70.9 ± 3.5	70.6 ± 3.2
E		✓	✓	87.3 ± 1.2	71.1 ± 1.5	71.2 ± 1.2
F	✓	✓	✓	**88.4 ± 1.3**	**74.0 ± 0.9**	**73.7 ± 0.6**

C. Ablation Study

To systematically evaluate the effectiveness of each component in our proposed ARGUS framework, we conducted comprehensive ablation experiments on DTH-ICC cohort(See Tab. II). Starting from a baseline MIL model, we observed that the addition of the Hierarchical FoVs Alignment (HFA) strategy consistently improved performance across both datasets, demonstrating its ability to capture complementary contextual cues. Incorporating micro-level geometry features(MGF) improved the model's ability to capture fine-grained structure. Moreover, integrating the Geometry Prior Guided Attention (GPGF) strategy proved critical for fusing multi-scale and geometry-aware representations. The full model, which integrates HFA, MGF, and GPGF, significantly outperformed models missing one or more modules, confirming that each module and their synergy enhance ARGUS's overall performance.

IV. CONCLUSION

In this paper, we propose **A** hie**R**archical **G**eometry-g**U**ided tran**S**former (ARGUS) to comprehensively model the macro–meso–micro hierarchical interactions within WSIs for the histological subtyping of primary liver cancer. Furthermore, by leveraging geometrical prior feature and FoVs histological feature, ARGUS effectively capture complementary morphological and cellular-level patterns. Extensive experiments conducted on both public and private cohorts demonstrate the effectiveness of our proposed ARGUS framework.

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